Computer Networks Laboratory
NS2 Simulator Introduction and TCL Script Execution Guide

1. Introduction to Network Simulator 2 (NS2)

Network Simulator 2 (NS2) is an open-source discrete event simulator used for computer network experiments. It helps students understand topology creation, packet transmission, routing, protocols, traffic generation and performance analysis.

NS2 uses TCL scripts to define:
- Nodes
- Links
- Protocols
- Traffic sources
- Simulation events
- Output files

2. NS2 Simulation Workflow

TCL Script
    |
    v
NS2 Simulation Engine
    |
    v
Trace File (.tr)
    |
    v
NAM Visualization (.nam)
    |
    v
Network Analysis

3. LAN Representation in NS2

LAN consists of:
- Nodes: Computers/end devices
- Links: Communication paths
- Queue: Packet storage mechanism
- Protocols: TCP/UDP
- Applications: FTP, CBR etc.

4. NS2 Installation (Ubuntu)

sudo apt update
sudo apt install ns2
sudo apt install nam

Check installation:

ns

5. TCL Script Execution Procedure

Step 1:
Create folder

mkdir ns2_lab
cd ns2_lab

Step 2:
Create TCL file

nano sample.tcl

Step 3:
Write TCL program and save.

CTRL+O
ENTER
CTRL+X

Step 4:
Execute

ns sample.tcl

Step 5:
Open NAM file

nam output.nam

6. Basic TCL Program Structure

Simulator creation
Node creation
Link creation
Protocol configuration
Traffic generation
Event scheduling
Simulation execution

7. **Simple Experiment 1:** Two Node LAN

Objective:
Create two nodes and establish communication.

Commands:

set ns [new Simulator]

set n0 [$ns node]
set n1 [$ns node]

$ns duplex-link $n0 $n1 1Mb 10ms DropTail

$ns at 5.0 finish
$ns run

8. TCP Transaction Demonstration

Flow:

Sender Node
|
TCP Agent
|
FTP Application
|
TCP Sink
|
Receiver Node

Execution:

ns tcp_demo.tcl
nam tcp_demo.nam

9. UDP Transaction Demonstration

Flow:

Sender Node
|
UDP Agent
|
CBR Traffic
|

Null Agent
|
Receiver Node

Execution:

ns udp_demo.tcl
nam udp_demo.nam

10. Important NS2 Commands

Create simulator:
set ns [new Simulator]

Create node:
set n0 [$ns node]

Create link:
$ns duplex-link n0 n1 bandwidth delay queue

Run:
$ns run

11. First Laboratory Activity

Students should:

1. Verify NS2 installation.
2. Create a two-node topology.
3. Execute TCL script.
4. View NAM animation.
5. Observe packet transmission.

Learning Outcome:
Students understand how computer networks are modeled and simulated using NS2.

Experiment 2: Creating Two Nodes and Link

Objective

Create two nodes and establish a point-to-point connection.

File: `two_nodes.tcl`

```
# Create simulator
set ns [new Simulator]

# Create nodes
set n0 [$ns node]
set n1 [$ns node]

# Create link
$ns duplex-link $n0 $n1 1Mb 10ms DropTail

# Create NAM file
set nf [open two_nodes.nam w]
$ns namtrace-all $nf

# Finish procedure
proc finish {} {
   global ns nf
   $ns flush-trace
   close $nf
   exit 0
}

# Simulation end
$ns at 5.0 "finish"

# Run
$ns run
```

Commands:

ns two_nodes.tcl

nam two_nodes.nam

Experiment 6: Packet Loss Demonstration

```
set ns [new Simulator]
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
$ns duplex-link $n0 $n1 1Mb 10ms DropTail
$ns duplex-link $n1 $n2 1Mb 10ms DropTail
# Queue limit
$ns queue-limit $n0 $n1 5
proc finish {} {
global ns
$ns flush-trace
exit 0
}
$ns at 10.0 finish
$ns run
```